System Risk Assessment — AI Assistant Platform (pilot)

Larkspur Logistics, Inc. · System Risk Assessment · SRA-2026-05 v1.0

Field	Value
Document id	SRA-2026-05
Version	1.0
Classification	INTERNAL — CONFIDENTIAL (fictitious)
System	AI Assistant Platform
Owner team	Applied AI
Assessment date	2026-06-19
Next review	2026-09-19
Assessor	Security Operations (SecOps), with the Applied AI team
Exposure	internal
Data classification	Confidential

1. Purpose and scope

This is the first assessment of the AI Assistant Platform, a pilot that gives staff an internal chat assistant grounded in company documentation and automates a small number of back-office workflows. The platform is new, moves quickly and combines several young open-source components that have had frequent security releases, which is why its review interval is three months rather than six.

2. System overview

The platform serves two assistants: a documentation assistant that answers staff questions from Confluence content, and a workflow assistant that drafts customer-service replies and files them in Jira. Flows are built in Langflow by the Applied AI team; models are served on premises by vLLM (two GPU nodes) with Ollama used by the team for local experiments on a shared workstation; n8n automates the hand-offs between the assistants, Jira and the e-mail system; and an MCP server (MCP Atlassian) gives the assistants read and write access to Confluence and Jira. About 260 pilot users in customer service and operations use the assistants; the Applied AI team has 9 members.

Data classification is Confidential: the assistants read internal documentation and customer-service tickets, and the workflow assistant writes to Jira. No customer personal data is stored on the platform itself, but it passes through prompts and the vector store. Availability tier is Tier 3 (best effort during the pilot).

Staff authenticate to the assistants through Keycloak OIDC. The Langflow, n8n and MCP Atlassian administration interfaces are reachable from the Applied AI team's network and the VPN; the MCP Atlassian server authenticates callers with its own user tokens and holds an operator service account for Confluence and Jira as a fallback.

3. Architecture and hosts

All components run on the AI cluster, a four-node Kubernetes cluster with two GPU workers, on the internal segment. Langflow hosts the flows and exposes them to the assistant front end through its API; vLLM serves the open-weight chat and embedding models through an OpenAI-compatible API protected

by an API key; Qdrant holds the document embeddings; n8n runs the workflows; MCP Atlassian runs as a streamable-HTTP MCP server in the same namespace and is called by the Langflow flows. Ollama runs on the team's shared GPU workstation on the office network, not in the cluster, and is used to try models before they are promoted to vLLM.

Confluence content is synchronised nightly into Qdrant through an n8n workflow that reads pages through MCP Atlassian; the documentation assistant retrieves from Qdrant at query time. The workflow assistant's Jira writes go through MCP Atlassian with the operator service account when the calling user's own token is not available.

3.1 Host inventory

Hostname	IP address	Zone	Role	Operating system
k8s-ai-cp-01.corp.example	198.51.100.180	Internal (AI)	Kubernetes control plane (AI cluster)	Ubuntu 24.04 LTS
k8s-ai-w-01.corp.example	198.51.100.181	Internal (AI)	Kubernetes worker (Langflow	Ubuntu 24.04 LTS
k8s-ai-gpu-01.corp.example	198.51.100.182	Internal (AI)	GPU worker (vLLM)	Ubuntu 24.04 LTS
k8s-ai-gpu-02.corp.example	198.51.100.183	Internal (AI)	GPU worker (vLLM)	Ubuntu 24.04 LTS
ai-lab-01.corp.example	198.51.100.190	Office network (Applied AI)	Shared GPU workstation (Ollama)	Ubuntu 24.04 LTS
bastion-01.mgmt.corp.example	192.0.2.5	Management	SSH bastion (MFA)	Ubuntu 24.04 LTS

4. Software bill of materials

The components below were confirmed from package manifests and running processes during the assessment. Versions are as deployed on the assessment date.

Component	Vendor / project	Version	Role in the system	Deployed on
Langflow	Langflow project (DataStax / IBM)	1.9.1	Flow builder and API for the two assistants	AI cluster (namespace langflow)
vLLM	vLLM project	0.21.0	Model serving (chat and embedding models) with an OpenAI-compatible API	k8s-ai-gpu-01, k8s-ai-gpu-02
Ollama	Ollama Inc.	0.30.1	Local model experiments on the shared workstation	ai-lab-01
n8n	n8n GmbH	2.20.3	Workflow automation between the assistants, Jira and e-mail	AI cluster (namespace n8n)
MCP Atlassian	sooperset (open source)	0.21.4	MCP server giving the assistants read and write access to Confluence and Jira	AI cluster (namespace langflow)
Qdrant	Qdrant Solutions GmbH	1.14.1	Vector store for the documentation assistant	AI cluster (namespace langflow)
Kubernetes	Kubernetes project (CNCF)	1.33.2	AI cluster	k8s-ai-cp-01, k8s-ai-w-01, k8s-ai-gpu-01, k8s-ai-gpu-02

4.1 Component notes

Langflow 1.9.1

Langflow 1.9.1 runs with authentication enabled (AUTO_LOGIN disabled) and Keycloak in front of the UI; the API used by the assistant front end is reachable from the cluster and the pilot users' networks with per-flow API keys. The Applied AI team's nine members are Langflow users who can create and edit flows, which includes custom Python components. Secrets for vLLM, Qdrant and MCP Atlassian are stored in Langflow's global variables. The instance is upgraded when the team adopts a new release, roughly every six weeks.

vLLM 0.21.0

vLLM 0.21.0 serves two models behind its OpenAI-compatible server with --api-key set; the endpoints /v1/chat/completions, /v1/embeddings and /pooling are reachable from the AI cluster and from the Applied AI team's network. Structured-output requests with regular expressions are used by the workflow assistant. The GPU nodes are dedicated to vLLM.

Ollama 0.30.1

Ollama 0.30.1 runs as a system service on the shared GPU workstation on the office network, listening on all interfaces so that team members can use it from their laptops; the experimental agent mode with the Bash tool is enabled for two team members' experiments. Models are pulled from the public registry through the egress proxy. The workstation is not part of the cluster and is patched by the team rather than by the platform process.

n8n 2.20.3

n8n 2.20.3 (Enterprise trial licence) with Keycloak SSO; 14 users across Applied AI and customer-service operations can create or modify workflows. Workflows use the HTTP Request node extensively and the Git node for the flow-export backup to GitLab. Credentials for Jira, the e-mail relay and MCP Atlassian are stored in n8n's credential store; the Dynamic Credentials feature is enabled for the customer-service workflows.

MCP Atlassian 0.21.4

MCP Atlassian 0.21.4 runs in streamable-HTTP mode with per-user token authentication for interactive use and the operator service account (a Confluence and Jira account with write access to the customer-service projects) configured as the global credential for the n8n synchronisation workflow. The endpoint is reachable from the langflow and n8n namespaces and, for debugging, from the Applied AI team's network; there is no separate authentication boundary in front of it.

Qdrant 1.14.1

Qdrant 1.14.1 with an API key, reachable only inside the cluster. It holds embeddings of the Confluence spaces the documentation assistant is allowed to answer from.

Kubernetes 1.33.2

Deployed and operated by Platform Engineering with the same baseline as the platform cluster; the GPU operator is installed on the two GPU workers. Network policies are applied per namespace.

5. Network exposure

The platform is internal: nothing is reachable from the internet. The assistants are reachable by the pilot users through the VPN and office networks; the administration interfaces and the MCP endpoint by the

Applied AI team's network and the VPN; the model-serving API by the cluster and the team's network. The Ollama workstation listens on the office network.

Two exposure factors specific to this platform: the assistants read content that other people wrote (Confluence pages, Jira tickets, customer e-mails), which is a prompt-injection path into the tools the assistants can call; and several components give authenticated users the ability to run code (Langflow custom components, n8n workflows, vLLM and Ollama model loading).

5.1 Exposed endpoints

Port / protocol	Service	Reachable from	Authentication
443/tcp	Assistant front end	VPN and office networks (pilot users)	Keycloak OIDC (MFA)
443/tcp	Langflow UI and API	Applied AI network	Keycloak OIDC for the UI; per-flow API keys
8000/tcp	vLLM OpenAI-compatible API	AI cluster and Applied AI network	API key (--api-key)
11434/tcp	Ollama API	Office network (all interfaces)	None
5678/tcp	n8n UI and webhooks	Applied AI network	Keycloak SSO; webhook tokens
8080/tcp	MCP Atlassian (streamable HTTP)	langflow and n8n namespaces; Applied AI network	Per-user tokens; operator service account fallback
6333/tcp	Qdrant	AI cluster only	API key

6. Security controls

Control	Status	Last verified	Notes
Not reachable from the internet	In place	2026-06-10	No ingress; egress through the proxy for model and package downloads.
SSO with MFA for the UIs	In place	2026-06-10	Langflow
Authentication boundary in front of the MCP endpoint	Not in place	2026-06-10	Relies on MCP Atlassian's own token check (F-2026-052).
Network policies per namespace	In place	2026-06-10	Applied by Platform Engineering.
Prompt-injection controls on retrieved content	Planned	2026-06-10	Being designed by the Applied AI team (F-2026-053).
Weekly authenticated vulnerability scan; images scanned in Harbor	Partial	2026-06-14	The Ollama workstation is not in the scan scope (F-2026-051).
Centralised logging to the SIEM	Partial	2026-06-10	Langflow and n8n logs shipped; vLLM and MCP Atlassian logs not yet.

7. Open findings

Id	Severity	Finding	Owner	Due
F-2026-051	High	The Ollama workstation listens on all office-network interfaces without authentication and is outside the scan and patch process.	Applied AI	2026-08-15
F-2026-052	High	The MCP Atlassian endpoint has no independent authentication boundary and holds an operator service account with write access to Jira.	Applied AI	2026-08-31

Id	Severity	Finding	Owner	Due
F-2026-053	Medium	No prompt-injection controls on content the assistants retrieve before it reaches tool-calling flows.	Applied AI	2026-09-30
F-2026-054	Medium	Secrets for vLLM	Applied AI	2026-09-30

8. Ownership and contacts

Role	Name	Contact
Owner Team lead	Elena Petrova	elena.petrova@larkspur-logistics.example
Technical lead (flows and models)	Rafael Moreira	rafael.moreira@larkspur-logistics.example
Technical lead (automation and integrations)	Hannah Weiss	hannah.weiss@larkspur-logistics.example
SecOps liaison	Priya Nandakumar	secops@larkspur-logistics.example

Applied AI owns the platform's applications and the Ollama workstation; Platform Engineering operates the AI cluster. Because the platform is a pilot, SecOps reviews it quarterly and any expansion beyond the pilot user group requires a new assessment.

9. Assessment history

Date	Version	Summary
2026-06-19	1.0	Initial assessment of the pilot; four findings open

Appendix A. Service and port matrix

Host	Port	Service	Listens on	Purpose
AI cluster ingress	443	NGINX ingress controller	198.51.100.179	Front end
k8s-ai-gpu-01/02	8000	vLLM	198.51.100.182-183	Model serving
k8s-ai-w-01	8080	MCP Atlassian	cluster service	Confluence and Jira tools
k8s-ai-w-01	6333	Qdrant	cluster service	Vector store
ai-lab-01	11434	Ollama	198.51.100.190 (all interfaces)	Local experiments

Appendix B. Backup, recovery and change management

Backups: Langflow flows and n8n workflows are exported nightly to a GitLab repository by an n8n workflow; the Langflow and n8n databases are backed up nightly to the object store; Qdrant is rebuilt from Confluence by the synchronisation workflow and is not backed up; model weights are re-downloadable. No restore has been exercised yet (pilot).

Change management: the Applied AI team deploys flow and workflow changes directly during the pilot and records them on the team's change page; component upgrades go through the weekly CAB. The pilot's exit criteria include moving flow changes to the GitLab deployment pipeline.

Appendix C. Change log

Date	Change	Approved by
2026-04-03	Pilot platform deployed; Langflow 1.9.1	CAB
2026-05-06	MCP Atlassian 0.21.4 deployed; operator service account created.	CAB
2026-06-02	Ollama 0.30.1 installed on the shared workstation; agent mode enabled for two users.	Owner Team